

NBA ANALYTICS AUTOMATION PLATFORM

PROCESS OPTIMIZATION, CLOUD INTEGRATION & KPI REPORTING AUTOMATION

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TECHNOLOGY & PROCESS OPTIMIZATION

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EXECUTIVE SUMMARY

This project began as an NBA analytics dashboard designed to compare player salaries against on-court impact and identify high-value performers. As the project evolved, I expanded it into a broader technology and process optimization initiative focused on automating reporting workflows and improving operational efficiency.

To support scalable analytics and reduce manual processes, I designed and implemented an automated cloud-integrated pipeline using Python, Azure, and Power BI. The solution automated data ingestion, standardized transformation workflows, centralized processed datasets in cloud storage, and enabled KPI-driven reporting through automated dashboard refreshes.

EXECUTIVE SUMMARY (CONT.)

The project demonstrates competencies in:

- Process mapping & workflow optimization
- Requirements gathering & solution evaluation
- ETL automation & cloud orchestration
- Business intelligence & KPI reporting
- Systems integration & operational analytics

Although the dataset focused on NBA performance data, the workflows and architecture were designed to mirror real-world enterprise reporting and business technology environments.



IDENTIFYING WORKFLOW INEFFICIENCIES

The original analytics workflow relied heavily on repetitive manual processes to collect, clean, transform, and report NBA statistical data. While functional at a small scale, the process introduced several operational inefficiencies that limited reporting consistency, scalability, and refresh reliability.

Key Challenges Identified

- Repetitive manual data preparation
- Inconsistent dashboard refresh workflows
- Duplicate/traded-player record handling
- Fragmented reporting processes
- Limited scalability for larger datasets
- Slower reporting turnaround times
- Increased risk of human error

As the project expanded, it became clear that the workflow required a more scalable and automated solution capable of supporting standardized reporting, centralized data management, and repeatable analytics processes.



BUSINESS & FUNCTIONAL REQUIREMENTS GATHERED

- Automate data ingestion and refresh workflows
- Standardize data transformation processes
- Centralize processed datasets in cloud storage
- Support scalable KPI-driven reporting
- Reduce manual operational dependency
- Improve reporting consistency and reliability
- Support repeatable analytics workflows

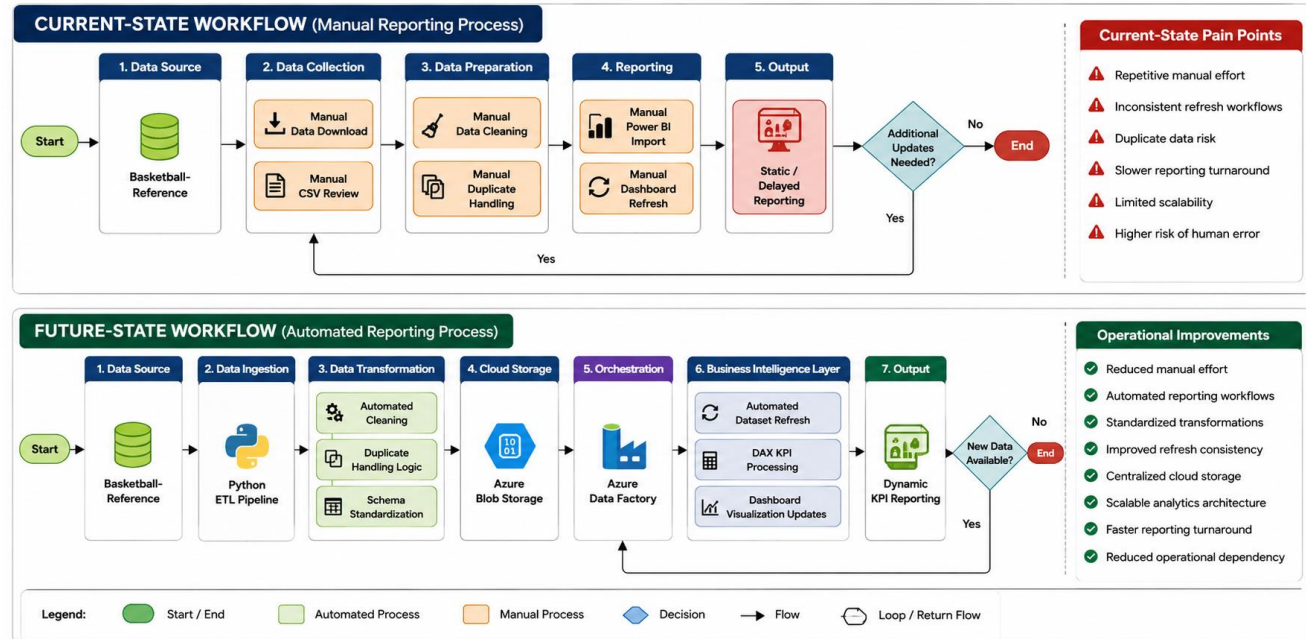
SAAS & SOLUTION EVALUATION

To support these requirements, multiple technologies and workflow approaches were evaluated based on scalability, automation capabilities, reporting integration, and operational fit.

This analysis ultimately led to the redesign of the workflow into a scalable cloud-integrated business intelligence pipeline.

Requirement	Selected Solution	Reason
Automated data ingestion	Python ETL	Flexible transformation logic
Centralized storage	Azure Blob Storage	Scalable cloud storage
Workflow orchestration	Azure Data Factory	Automated scheduling & pipeline control
KPI reporting	Power BI	Interactive analytics & DAX modeling
Data transformation	pandas + Power Query	Standardized transformation workflows

CURRENT-STATE VS FUTURE-STATE WORKFLOW

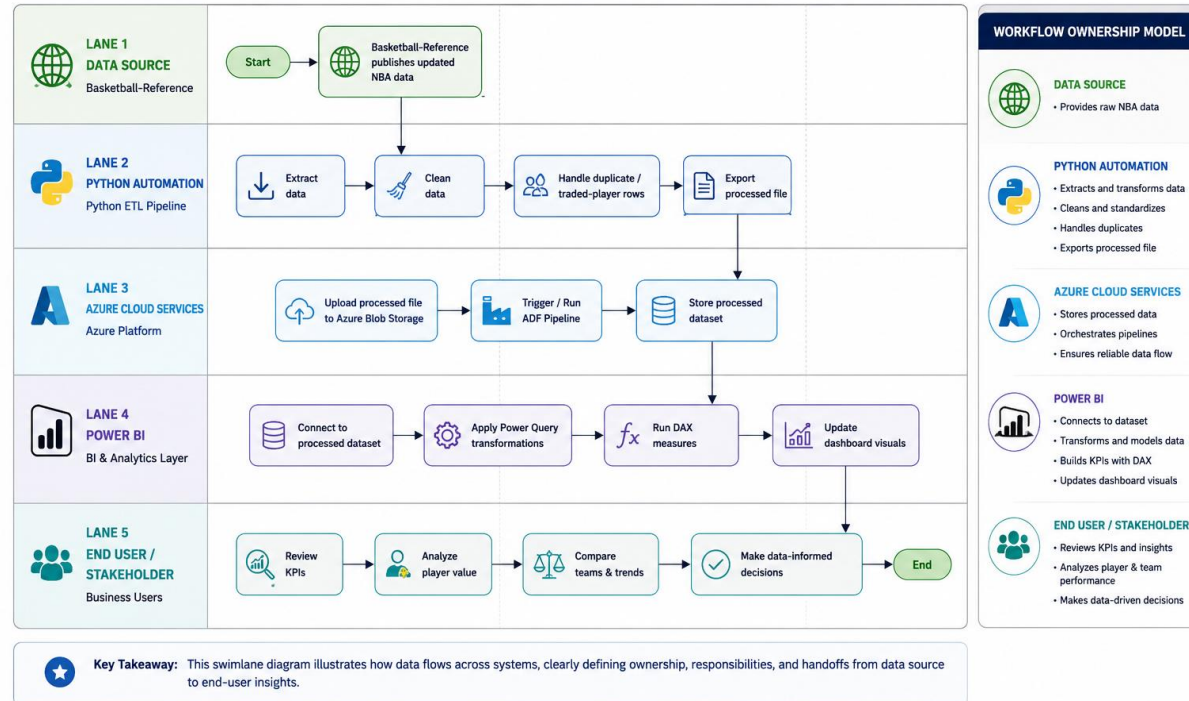


CROSS-FUNCTIONAL WORKFLOW OWNERSHIP

SWIMLANE DIAGRAM: NBA Analytics Workflow

Cross-Functional Workflow & Ownership

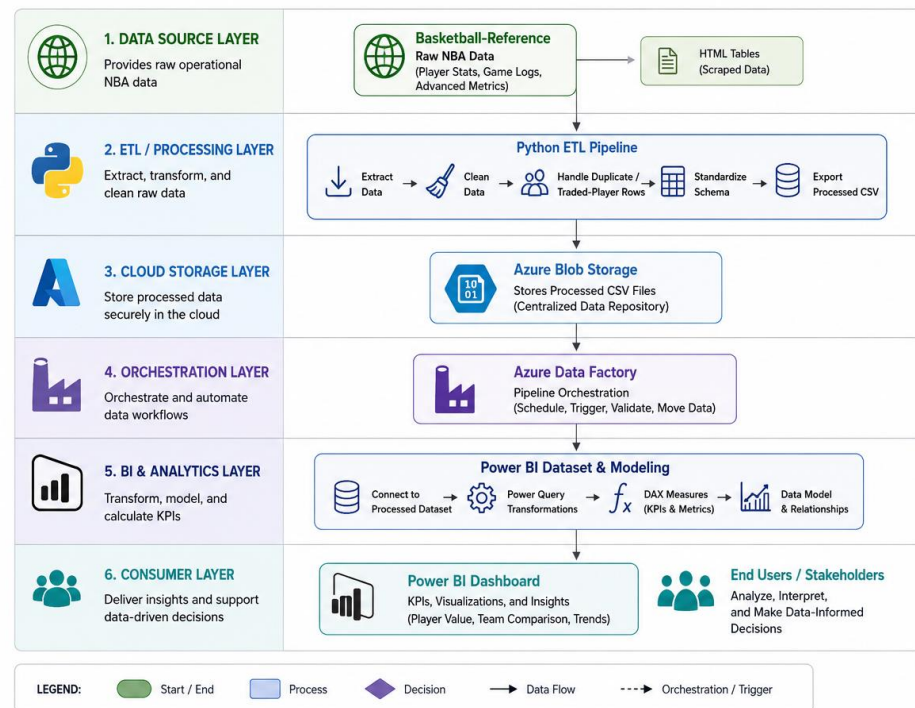
START / END PROCESS DECISION



CLOUD- INTEGRATED ANALYTICS ARCHITECTURE

SYSTEM ARCHITECTURE DIAGRAM: NBA Analytics Platform

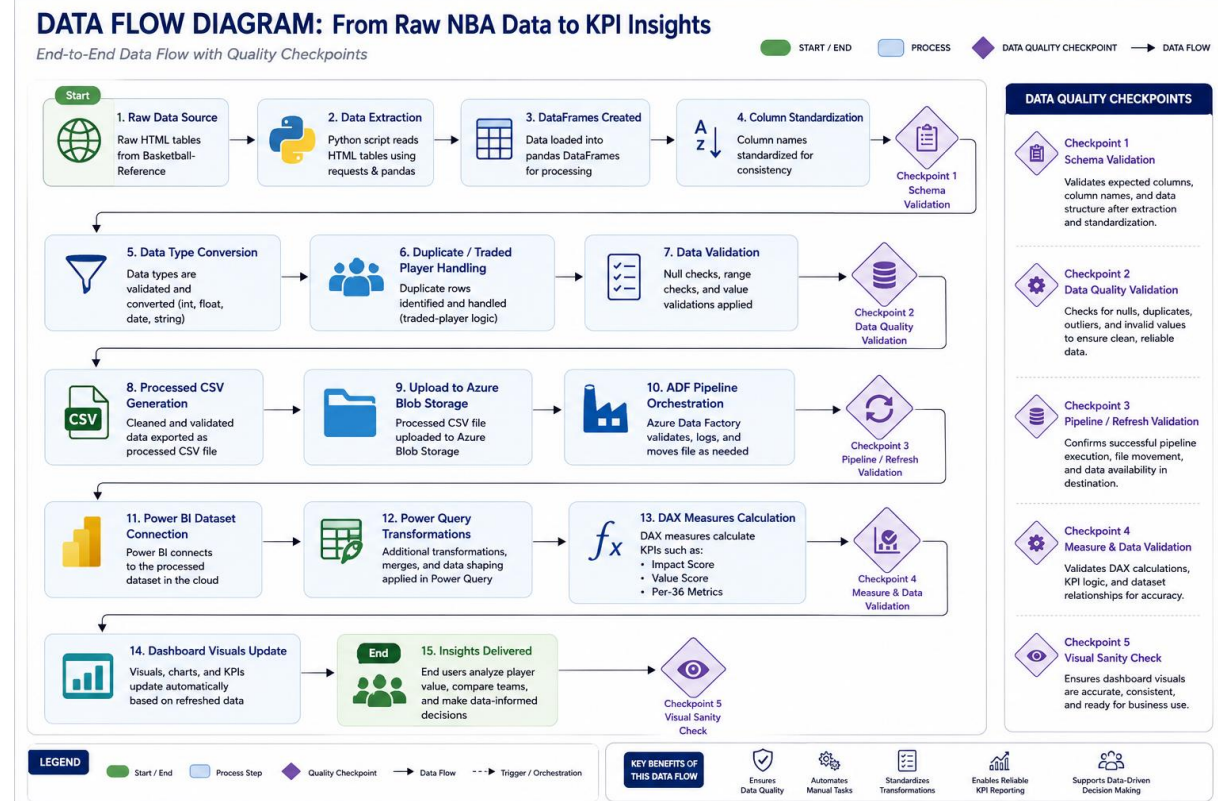
End-to-End Architecture for Automated NBA Data Pipeline and KPI Reporting



CORE ARCHITECTURE PRINCIPLES

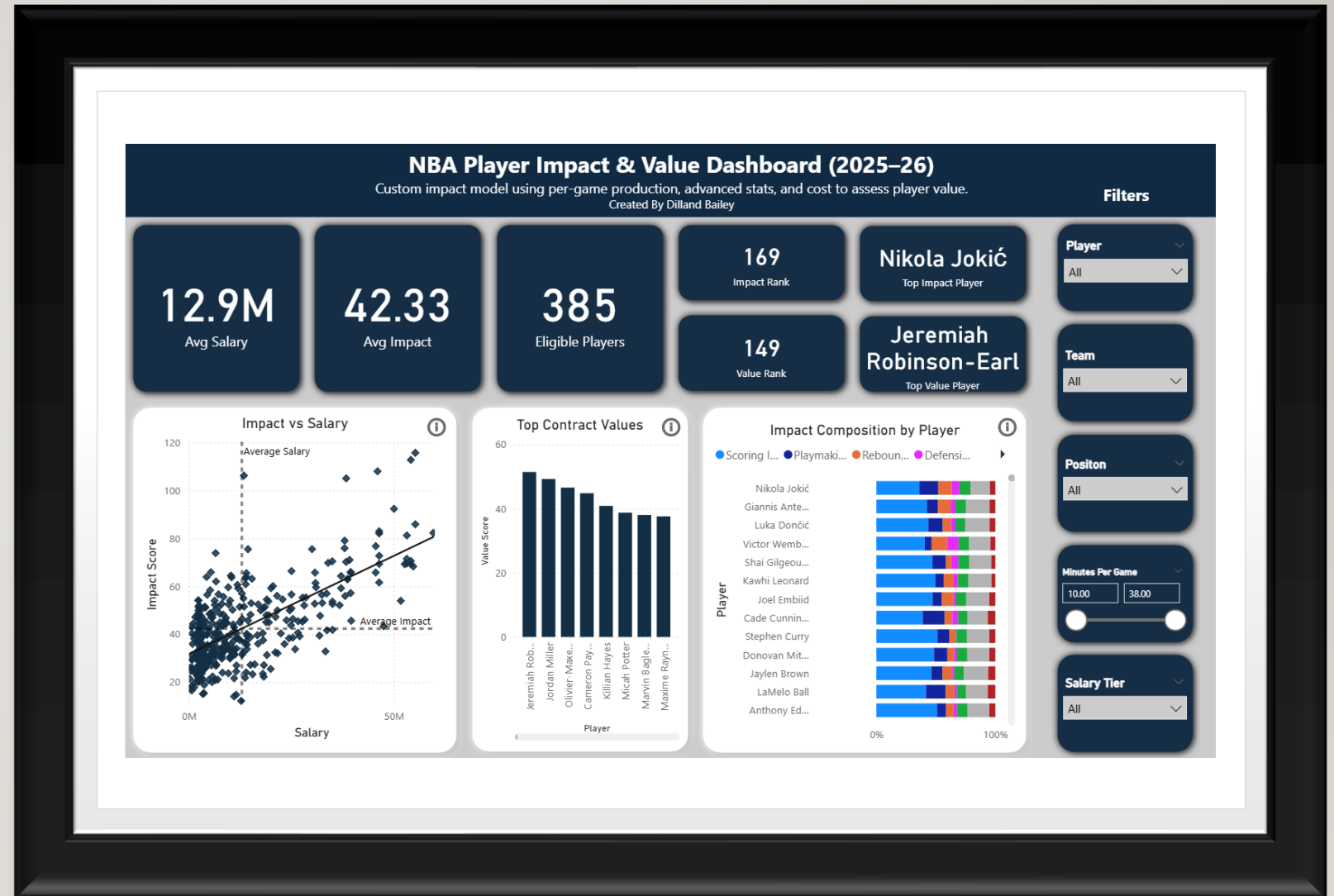
- Automated Ingestion**
Automates data extraction and pipeline execution
- Centralized Storage**
All processed data is securely stored in Azure Cloud
- Repeatable Transformation**
Consistent cleaning, validation, and transformation logic
- BI-Ready Dataset**
Clean, modeled, and optimized dataset for KPI calculation
- Scalable Reporting Layer**
Power BI dashboards deliver actionable insights at scale
- Reliable & Maintainable**
Modular architecture ensures reliability, scalability, and easy maintenance

END-TO-END DATA TRANSFORMATION LIFECYCLE



DASHBOARD & KPI REPORTING

[Live Link to Dashboard](#)



TECHNICAL CHALLENGES & SOLUTIONS

Challenge	Impact	Solution Implemented
Duplicate traded-player records (2TM/3TM)	Inconsistent KPI calculations and duplicate reporting	Implemented combined-row transformation logic to standardize player records
Manual dashboard refresh dependency	Repetitive operational effort and delayed reporting	Designed automated reporting refresh workflows
Dataset consistency across multiple sources	Schema mismatches and reporting inconsistencies	Applied standardized transformation and validation logic
Workflow orchestration & scheduling	Limited automation and process scalability	Implemented Azure Data Factory pipeline orchestration
Data validation and refresh accuracy	Risk of inaccurate KPI reporting	Added validation checkpoints and transformation reviews

BUSINESS VALUE & OPERATIONAL IMPACT

Operational Improvements

- Reduced repetitive manual effort
- Standardized reporting workflows
- Scalable analytics architecture
- Improved refresh consistency
- Centralized cloud-based data management
- Automated reporting lifecycle

Business Outcomes

- Improved reporting efficiency
- Faster analytics delivery
- Repeatable reporting processes
- Reduced operational dependency
- Improved data consistency and reliability
- Enhanced KPI-driven decision support

Organizational Impact

The redesigned workflow transformed a fragmented reporting process into a scalable cloud-integrated analytics platform capable of supporting repeatable and reliable business intelligence operations.



KEY TAKEAWAYS / WHY THIS MATTERS

Why This Matters

- This project demonstrates how analytics, automation, cloud infrastructure, and process optimization can be integrated into a scalable business intelligence workflow.
- Although the operational dataset focused on NBA performance data, the architecture and workflow strategies were designed to mirror real-world enterprise reporting and business technology environments.

Core Competencies Demonstrated

- Analytics engineering
- Workflow optimization
- Cloud integration
- Business intelligence reporting
- ETL automation
- Systems thinking
- Process mapping
- Operational analytics
- Requirements gathering
- Technology evaluation

PROJECT IMPACT & PROFESSIONAL ALIGNMENT

Technologies Utilized

- Miro
- Python
- pandas
- Azure Blob Storage
- Azure Data Factory
- Power BI
- DAX
- Power Query

Core Business Concepts Applied

- Requirements gathering
- Process mapping
- SaaS / technology evaluation
- Workflow redesign
- Reporting automation
- KPI modeling
- Operational analytics